UniRoute — OpenAPI 3.0 Specification

Version: 1.0.0 | **Format:** YAML | **用途:** 前后端开发契约

```yaml
openapi: 3.0.3
info:
  title: UniRoute API
  description: |
    UniRoute 企业级 AI 模型管控平台 API。

    **Gateway 端点**兼容 OpenAI API 格式。
    **Admin 端点**用于平台管理。

    认证方式:
    - Gateway: `Authorization: Bearer ur-XXXX-...`
    - Admin: `Authorization: Bearer <JWT>`
  version: 1.0.0

servers:
  - url: http://localhost:8080
    description: 本地开发

tags:
  - name: Gateway
    description: OpenAI 兼容网关端点
  - name: Auth
    description: 管理员认证
  - name: Admin - Organizations
    description: 组织管理
  - name: Admin - Users
    description: 用户管理
  - name: Admin - Channels
    description: 渠道管理
  - name: Admin - API Keys
    description: API Key 管理
  - name: Admin - Wallets
    description: 钱包管理
  - name: Admin - Prices
    description: 模型价格管理
  - name: Admin - Logs
    description: 调用日志
  - name: Admin - Dashboard
    description: 仪表盘
  - name: Admin - Settings
    description: 系统设置
  - name: Health
    description: 健康检查

paths:
```

```
  # ========================================
  # Gateway 端点 (OpenAI 兼容)
  # ========================================
  /v1/chat/completions:
    post:
      tags: [Gateway]
      summary: 聊天补全
      description: |
        OpenAI 兼容的聊天补全接口。支持流式 (SSE) 和非流式。
        响应体包含 `_uniroute` 扩展字段。
      parameters:
        - $ref: '#/components/parameters/XRequestId'
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ChatCompletionRequest'
      responses:
        '200':
          description: 成功
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ChatCompletionResponse'
            text/event-stream:
              schema:
                type: string
                description: SSE 流, 每个 data 行是 JSON, 结束标记 data:
[DONE]
        '401':
          $ref: '#/components/responses/Unauthorized'
        '402':
          $ref: '#/components/responses/InsufficientBalance'
        '403':
          $ref: '#/components/responses/Forbidden'
        '429':
          $ref: '#/components/responses/RateLimited'
        '502':
          $ref: '#/components/responses/UpstreamError'

  /v1/embeddings:
    post:
      tags: [Gateway]
      summary: 文本嵌入
      parameters:
        - $ref: '#/components/parameters/XRequestId'
      requestBody:
        required: true
        content:
          application/json:
            schema:
              type: object
              required: [model, input]
              properties:
                model:
                  type: string
                  example: qwen3.7-max
                input:
                  oneOf:
```

```
                  - type: string
                  - type: array
                    items:
                      type: string
              encoding_format:
                type: string
                enum: [float, base64]
                default: float
    responses:
      '200':
        description: 成功
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/EmbeddingResponse'
      '401':
        $ref: '#/components/responses/Unauthorized'
      '402':
        $ref: '#/components/responses/InsufficientBalance'

/v1/models:
  get:
    tags: [Gateway]
    summary: 模型列表
    responses:
      '200':
        description: 成功
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ModelListResponse'

# ========================================
# 认证
# ========================================
/api/v1/auth/login:
  post:
    tags: [Auth]
    summary: 管理员登录
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            required: [username, password]
            properties:
              username:
                type: string
              password:
                type: string
    responses:
      '200':
        description: 登录成功
        content:
          application/json:
            schema:
              type: object
              properties:
                token:
```

```
                  type: string
                  description: JWT token
                expires_in:
                  type: integer
                  description: 过期时间（秒）
                user:
                  $ref: '#/components/schemas/User'
        '401':
          description: 用户名或密码错误

  # ========================================
  # 组织管理
  # ========================================
  /api/v1/admin/orgs:
    get:
      tags: [Admin - Organizations]
      summary: 组织列表
      security:
        - bearerAuth: []
      parameters:
        - $ref: '#/components/parameters/Page'
        - $ref: '#/components/parameters/PageSize'
      responses:
        '200':
          description: 成功
          content:
            application/json:
              schema:
                type: object
                properties:
                  data:
                    type: array
                    items:
                      $ref: '#/components/schemas/Organization'
                  total:
                    type: integer
    post:
      tags: [Admin - Organizations]
      summary: 创建组织
      security:
        - bearerAuth: []
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/OrganizationCreate'
      responses:
        '201':
          description: 创建成功
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Organization'

  /api/v1/admin/orgs/{id}:
    get:
      tags: [Admin - Organizations]
      summary: 获取组织详情
      security:
```

```
        - bearerAuth: []
      parameters:
        - $ref: '#/components/parameters/Id'
      responses:
        '200':
          description: 成功
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Organization'
    put:
      tags: [Admin - Organizations]
      summary: 更新组织
      security:
        - bearerAuth: []
      parameters:
        - $ref: '#/components/parameters/Id'
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/OrganizationUpdate'
      responses:
        '200':
          description: 更新成功
    delete:
      tags: [Admin - Organizations]
      summary: 删除组织（软删除）
      security:
        - bearerAuth: []
      parameters:
        - $ref: '#/components/parameters/Id'
      responses:
        '204':
          description: 删除成功

  # ========================================
  # 渠道管理
  # ========================================
  /api/v1/admin/channels:
    get:
      tags: [Admin - Channels]
      summary: 渠道列表
      security:
        - bearerAuth: []
      parameters:
        - $ref: '#/components/parameters/Page'
        - $ref: '#/components/parameters/PageSize'
        - name: status
          in: query
          schema:
            type: string
            enum: [active, deprecated, disabled]
      responses:
        '200':
          description: 成功
          content:
            application/json:
              schema:
```

```
                type: object
                properties:
                  data:
                    type: array
                    items:
                      $ref: '#/components/schemas/Channel'
                  total:
                    type: integer
    post:
      tags: [Admin - Channels]
      summary: 创建渠道
      security:
        - bearerAuth: []
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ChannelCreate'
      responses:
        '201':
          description: 创建成功

  /api/v1/admin/channels/{id}:
    get:
      tags: [Admin - Channels]
      summary: 渠道详情
      security:
        - bearerAuth: []
      parameters:
        - $ref: '#/components/parameters/Id'
      responses:
        '200':
          description: 成功
    put:
      tags: [Admin - Channels]
      summary: 更新渠道
      security:
        - bearerAuth: []
      parameters:
        - $ref: '#/components/parameters/Id'
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ChannelUpdate'
      responses:
        '200':
          description: 更新成功
    delete:
      tags: [Admin - Channels]
      summary: 删除渠道
      security:
        - bearerAuth: []
      parameters:
        - $ref: '#/components/parameters/Id'
      responses:
        '204':
          description: 删除成功
```

```
  /api/v1/admin/channels/{id}/test:
    post:
      tags: [Admin - Channels]
      summary: 测试渠道连通性
      description: |
        向渠道发送测试请求，返回响应内容、费用、延迟、curl 示例。
      security:
        - bearerAuth: []
      parameters:
        - $ref: '#/components/parameters/Id'
      requestBody:
        content:
          application/json:
            schema:
              type: object
              properties:
                model:
                  type: string
                  description: 测试用模型名
                message:
                  type: string
                  default: "Hello, this is a test."
      responses:
        '200':
          description: 测试结果
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ChannelTestResult'

  # ========================================
  # API Key 管理
  # ========================================
  /api/v1/admin/keys:
    get:
      tags: [Admin - API Keys]
      summary: API Key 列表
      security:
        - bearerAuth: []
      parameters:
        - $ref: '#/components/parameters/Page'
        - $ref: '#/components/parameters/PageSize'
        - name: org_id
          in: query
          schema:
            type: integer
      responses:
        '200':
          description: 成功
          content:
            application/json:
              schema:
                type: object
                properties:
                  data:
                    type: array
                    items:
                      $ref: '#/components/schemas/ApiKey'
                  total:
```

```
                    type: integer
    post:
      tags: [Admin - API Keys]
      summary: 创建 API Key
      description: 返回完整 Key（只显示一次）。
      security:
        - bearerAuth: []
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ApiKeyCreate'
      responses:
        '201':
          description: 创建成功
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/ApiKeyCreated'

  /api/v1/admin/keys/{id}:
    get:
      tags: [Admin - API Keys]
      summary: API Key 详情（不含明文）
      security:
        - bearerAuth: []
      parameters:
        - $ref: '#/components/parameters/Id'
      responses:
        '200':
          description: 成功
    put:
      tags: [Admin - API Keys]
      summary: 更新 API Key
      security:
        - bearerAuth: []
      parameters:
        - $ref: '#/components/parameters/Id'
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ApiKeyUpdate'
      responses:
        '200':
          description: 更新成功
    delete:
      tags: [Admin - API Keys]
      summary: 吊销 API Key
      security:
        - bearerAuth: []
      parameters:
        - $ref: '#/components/parameters/Id'
      responses:
        '204':
          description: 吊销成功

  # ========================================
```

```yaml
  # 钱包管理
  # ========================================
  /api/v1/admin/wallets/{org_id}:
    get:
      tags: [Admin - Wallets]
      summary: 获取组织钱包
      security:
        - bearerAuth: []
      parameters:
        - name: org_id
          in: path
          required: true
          schema:
            type: integer
      responses:
        '200':
          description: 成功
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Wallet'

  /api/v1/admin/wallets/{org_id}/recharge:
    post:
      tags: [Admin - Wallets]
      summary: 充值
      security:
        - bearerAuth: []
      parameters:
        - name: org_id
          in: path
          required: true
          schema:
            type: integer
      requestBody:
        required: true
        content:
          application/json:
            schema:
              type: object
              required: [amount, client_request_id]
              properties:
                amount:
                  type: integer
                  description: 充值金额（分）
                client_request_id:
                  type: string
                  description: 幂等 key
                note:
                  type: string
      responses:
        '200':
          description: 充值成功
        '409':
          description: 重复请求

  /api/v1/admin/wallets/{org_id}/adjust:
    post:
      tags: [Admin - Wallets]
      summary: 管理员调账
```

```
      security:
        - bearerAuth: []
      parameters:
        - name: org_id
          in: path
          required: true
          schema:
            type: integer
      requestBody:
        required: true
        content:
          application/json:
            schema:
              type: object
              required: [amount, client_request_id]
              properties:
                amount:
                  type: integer
                  description: 调账金额（分，正数增加，负数减少）
                client_request_id:
                  type: string
                reason:
                  type: string
      responses:
        '200':
          description: 调账成功

  /api/v1/admin/wallets/{org_id}/grant:
    post:
      tags: [Admin - Wallets]
      summary: 赠送额度
      security:
        - bearerAuth: []
      parameters:
        - name: org_id
          in: path
          required: true
          schema:
            type: integer
      requestBody:
        required: true
        content:
          application/json:
            schema:
              type: object
              required: [amount, client_request_id]
              properties:
                amount:
                  type: integer
                  description: 赠送金额（分）
                client_request_id:
                  type: string
                expires_at:
                  type: string
                  format: date-time
                  description: 过期时间
                reason:
                  type: string
      responses:
        '200':
```

```
        description: 赠送成功

  # ========================================
  # 模型价格管理
  # ========================================
  /api/v1/admin/prices:
    get:
      tags: [Admin - Prices]
      summary: 模型价格列表
      security:
        - bearerAuth: []
      responses:
        '200':
          description: 成功
          content:
            application/json:
              schema:
                type: object
                properties:
                  data:
                    type: array
                    items:
                      $ref: '#/components/schemas/ModelPrice'
    post:
      tags: [Admin - Prices]
      summary: 创建/更新模型价格
      security:
        - bearerAuth: []
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ModelPriceCreate'
      responses:
        '200':
          description: 成功

  /api/v1/admin/prices/overrides:
    get:
      tags: [Admin - Prices]
      summary: 组织级定价覆盖列表
      security:
        - bearerAuth: []
      parameters:
        - name: org_id
          in: query
          schema:
            type: integer
      responses:
        '200':
          description: 成功
    post:
      tags: [Admin - Prices]
      summary: 创建/更新组织级覆盖
      security:
        - bearerAuth: []
      requestBody:
        required: true
        content:
```

```
          application/json:
            schema:
              $ref: '#/components/schemas/PriceOverride'
      responses:
        '200':
          description: 成功

  # ========================================
  # 调用日志
  # ========================================
  /api/v1/admin/logs:
    get:
      tags: [Admin - Logs]
      summary: 调用日志列表
      security:
        - bearerAuth: []
      parameters:
        - $ref: '#/components/parameters/Page'
        - $ref: '#/components/parameters/PageSize'
        - name: org_id
          in: query
          schema:
            type: integer
        - name: model
          in: query
          schema:
            type: string
        - name: status
          in: query
          schema:
            type: string
            enum: [success, error]
        - name: start_time
          in: query
          schema:
            type: string
            format: date-time
        - name: end_time
          in: query
          schema:
            type: string
            format: date-time
      responses:
        '200':
          description: 成功
          content:
            application/json:
              schema:
                type: object
                properties:
                  data:
                    type: array
                    items:
                      $ref: '#/components/schemas/CallLog'
                  total:
                    type: integer

  /api/v1/admin/logs/export:
    get:
      tags: [Admin - Logs]
```

```
      summary: 导出调用日志 CSV
      security:
        - bearerAuth: []
      parameters:
        - name: org_id
          in: query
          schema:
            type: integer
        - name: start_time
          in: query
          schema:
            type: string
            format: date-time
        - name: end_time
          in: query
          schema:
            type: string
            format: date-time
      responses:
        '200':
          description: CSV 文件
          content:
            text/csv:
              schema:
                type: string

  # ========================================
  # Dashboard
  # ========================================
  /api/v1/admin/dashboard:
    get:
      tags: [Admin - Dashboard]
      summary: 仪表盘统计数据
      security:
        - bearerAuth: []
      parameters:
        - name: period
          in: query
          schema:
            type: string
            enum: [today, week, month]
            default: today
      responses:
        '200':
          description: 成功
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Dashboard'

  # ========================================
  # 系统设置
  # ========================================
  /api/v1/admin/settings:
    get:
      tags: [Admin - Settings]
      summary: 获取所有设置
      security:
        - bearerAuth: []
      responses:
```

```
        '200':
          description: 成功
    put:
      tags: [Admin - Settings]
      summary: 更新设置
      security:
        - bearerAuth: []
      requestBody:
        required: true
        content:
          application/json:
            schema:
              type: object
              additionalProperties: true
      responses:
        '200':
          description: 更新成功

  # ========================================
  # 健康检查
  # ========================================
  /health/live:
    get:
      tags: [Health]
      summary: 存活检查
      responses:
        '200':
          description: 服务存活
          content:
            application/json:
              schema:
                type: object
                properties:
                  status:
                    type: string
                    example: ok

  /health/ready:
    get:
      tags: [Health]
      summary: 就绪检查
      responses:
        '200':
          description: 服务就绪
          content:
            application/json:
              schema:
                type: object
                properties:
                  status:
                    type: string
                    example: ok
                  db:
                    type: string
                    example: ok
                  redis:
                    type: string
                    example: ok
        '503':
          description: 服务不可用
```

```yaml
# ========================================
# Components
# ========================================
components:
  securitySchemes:
    bearerAuth:
      type: http
      scheme: bearer
      bearerFormat: JWT

  parameters:
    Id:
      name: id
      in: path
      required: true
      schema:
        type: integer
    Page:
      name: page
      in: query
      schema:
        type: integer
        default: 1
    PageSize:
      name: page_size
      in: query
      schema:
        type: integer
        default: 20
        maximum: 100
    XRequestId:
      name: X-Request-Id
      in: header
      description: 幂等 key (UUID v4)
      schema:
        type: string
        format: uuid

  schemas:
    # Gateway
    ChatCompletionRequest:
      type: object
      required: [model, messages]
      properties:
        model:
          type: string
          example: deepseek-v4-pro
        messages:
          type: array
          items:
            type: object
            properties:
              role:
                type: string
                enum: [system, user, assistant]
              content:
                type: string
        stream:
          type: boolean
```

```
          default: false
        max_tokens:
          type: integer
        temperature:
          type: number
        top_p:
          type: number

    ChatCompletionResponse:
      type: object
      properties:
        id:
          type: string
        object:
          type: string
        created:
          type: integer
        model:
          type: string
        choices:
          type: array
          items:
            type: object
            properties:
              index:
                type: integer
              message:
                type: object
                properties:
                  role:
                    type: string
                  content:
                    type: string
              finish_reason:
                type: string
        usage:
          type: object
          properties:
            prompt_tokens:
              type: integer
            completion_tokens:
              type: integer
            total_tokens:
              type: integer
        _uniroute:
          type: object
          properties:
            request_id:
              type: string
            cost:
              type: object
              properties:
                amount:
                  type: number
                currency:
                  type: string
            provider:
              type: string
            channel:
              type: string
```

```
        ttft_ms:
          type: integer
        cache_hit:
          type: boolean
        fallback:
          type: boolean

    EmbeddingResponse:
      type: object
      properties:
        object:
          type: string
        data:
          type: array
          items:
            type: object
            properties:
              object:
                type: string
              embedding:
                type: array
                items:
                  type: number
              index:
                type: integer
        model:
          type: string
        usage:
          type: object
          properties:
            prompt_tokens:
              type: integer
            total_tokens:
              type: integer

    ModelListResponse:
      type: object
      properties:
        object:
          type: string
        data:
          type: array
          items:
            type: object
            properties:
              id:
                type: string
              object:
                type: string
              created:
                type: integer
              owned_by:
                type: string

    # Admin
    Organization:
      type: object
      properties:
        id:
          type: integer
```

```
        name:
          type: string
        status:
          type: string
          enum: [active, suspended, banned]
        org_type:
          type: string
          enum: [customer, partner]
        settings:
          type: object
        created_at:
          type: string
          format: date-time

    OrganizationCreate:
      type: object
      required: [name]
      properties:
        name:
          type: string
        org_type:
          type: string
          enum: [customer, partner]
          default: customer
        settings:
          type: object

    OrganizationUpdate:
      type: object
      properties:
        name:
          type: string
        status:
          type: string
          enum: [active, suspended, banned]
        settings:
          type: object

    Channel:
      type: object
      properties:
        id:
          type: integer
        name:
          type: string
        provider:
          type: string
        base_url:
          type: string
        models:
          type: string
          description: 逗号分隔的模型列表
        model_mapping:
          type: object
          description: 上游模型名映射
        max_context:
          type: integer
        default_max_output:
          type: integer
        priority:
```

```
          type: integer
        rpm_limit:
          type: integer
        tpm_limit:
          type: integer
        status:
          type: string
          enum: [active, deprecated, disabled]
        circuit_state:
          type: string
          enum: [healthy, open, half_open]
        keys:
          type: array
          items:
            type: object
            properties:
              id:
                type: integer
              status:
                type: string
              last_used_at:
                type: string
                format: date-time
        created_at:
          type: string
          format: date-time

    ChannelCreate:
      type: object
      required: [name, provider, base_url, models]
      properties:
        name:
          type: string
        provider:
          type: string
        base_url:
          type: string
        models:
          type: string
        model_mapping:
          type: object
        max_context:
          type: integer
          default: 128000
        default_max_output:
          type: integer
          default: 4096
        priority:
          type: integer
          default: 100
        rpm_limit:
          type: integer
          default: 600
        tpm_limit:
          type: integer
          default: 100000
        keys:
          type: array
          items:
            type: object
```

```
            properties:
              key:
                type: string
                description: 上游 API Key（明文，服务端加密存储）

    ChannelUpdate:
      type: object
      properties:
        name:
          type: string
        base_url:
          type: string
        models:
          type: string
        model_mapping:
          type: object
        max_context:
          type: integer
        default_max_output:
          type: integer
        priority:
          type: integer
        rpm_limit:
          type: integer
        tpm_limit:
          type: integer
        status:
          type: string
          enum: [active, deprecated, disabled]

    ChannelTestResult:
      type: object
      properties:
        success:
          type: boolean
        response:
          type: string
          description: 上游响应内容
        cost:
          type: object
          properties:
            amount:
              type: number
            currency:
              type: string
        latency:
          type: object
          properties:
            ttft_ms:
              type: integer
            total_ms:
              type: integer
        curl_example:
          type: string
        error:
          type: object
          properties:
            message:
              type: string
            hint:
```

```
            type: string
          upstream_error:
            type: string

    ApiKey:
      type: object
      properties:
        id:
          type: integer
        org_id:
          type: integer
        name:
          type: string
        key_prefix:
          type: string
        ip_whitelist:
          type: string
        allowed_models:
          type: string
        rpm_limit:
          type: integer
        qps_limit:
          type: integer
        budget_limit:
          type: integer
        status:
          type: string
          enum: [active, disabled, revoked]
        created_at:
          type: string
          format: date-time

    ApiKeyCreate:
      type: object
      required: [org_id]
      properties:
        org_id:
          type: integer
        name:
          type: string
        ip_whitelist:
          type: string
          description: 逗号分隔的 IP 列表
        allowed_models:
          type: string
          description: 逗号分隔的模型列表，空表示不限制
        rpm_limit:
          type: integer
          default: 60
        qps_limit:
          type: integer
          default: 10
        budget_limit:
          type: integer
          description: 预算上限（分）

    ApiKeyCreated:
      type: object
      properties:
        id:
```

```
          type: integer
        key:
          type: string
          description: 完整 Key（只显示一次）
        key_prefix:
          type: string
        message:
          type: string

    ApiKeyUpdate:
      type: object
      properties:
        name:
          type: string
        ip_whitelist:
          type: string
        allowed_models:
          type: string
        rpm_limit:
          type: integer
        qps_limit:
          type: integer
        budget_limit:
          type: integer
        status:
          type: string
          enum: [active, disabled]

    Wallet:
      type: object
      properties:
        org_id:
          type: integer
        balance:
          type: integer
          description: 充值余额（分）
        granted_balance:
          type: integer
          description: 赠送余额（分）
        frozen:
          type: integer
          description: 冻结金额（分）
        total_spent:
          type: integer
          description: 累计消费（分）
        updated_at:
          type: string
          format: date-time

    ModelPrice:
      type: object
      properties:
        id:
          type: integer
        model_name:
          type: string
        provider:
          type: string
        input_price_cents:
          type: integer
```

```
        description: 输入价格（分/百万 Token）
      output_price_cents:
        type: integer
        description: 输出价格（分/百万 Token）
      max_context:
        type: integer
      default_max_output:
        type: integer

  ModelPriceCreate:
    type: object
    required: [model_name, input_price_cents, output_price_cents]
    properties:
      model_name:
        type: string
      provider:
        type: string
      input_price_cents:
        type: integer
      output_price_cents:
        type: integer
      max_context:
        type: integer
      default_max_output:
        type: integer

  PriceOverride:
    type: object
    required: [org_id, model_name, multiplier]
    properties:
      org_id:
        type: integer
      model_name:
        type: string
      multiplier:
        type: number
        description: 价格倍数（如 1.15 = 加价 15%）

  CallLog:
    type: object
    properties:
      id:
        type: integer
      request_id:
        type: string
      org_id:
        type: integer
      api_key_id:
        type: integer
      model:
        type: string
      provider:
        type: string
      channel:
        type: string
      prompt_tokens:
        type: integer
      completion_tokens:
        type: integer
      total_tokens:
```

```
          type: integer
        cost_cents:
          type: integer
        latency_ms:
          type: integer
        ttft_ms:
          type: integer
        status_code:
          type: integer
        upstream_status_code:
          type: integer
        upstream_error:
          type: string
        usage_source:
          type: string
          enum: [upstream, estimated]
        is_stream:
          type: boolean
        created_at:
          type: string
          format: date-time

    Dashboard:
      type: object
      properties:
        period:
          type: string
        total_requests:
          type: integer
        total_cost_cents:
          type: integer
        active_keys:
          type: integer
        active_orgs:
          type: integer
        top_models:
          type: array
          items:
            type: object
            properties:
              model:
                type: string
              requests:
                type: integer
              cost_cents:
                type: integer
        cost_trend:
          type: array
          items:
            type: object
            properties:
              date:
                type: string
              cost_cents:
                type: integer

    User:
      type: object
      properties:
        id:
```

```
          type: integer
        org_id:
          type: integer
        username:
          type: string
        role:
          type: string
          enum: [platform_admin, org_admin, org_viewer]
        status:
          type: string

    Error:
      type: object
      properties:
        error:
          type: object
          properties:
            message:
              type: string
            type:
              type: string
            code:
              type: string
            param:
              type: string
              nullable: true

  responses:
    Unauthorized:
      description: 认证失败
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
    InsufficientBalance:
      description: 余额不足
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
    Forbidden:
      description: 权限不足
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
    RateLimited:
      description: 限流
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
    UpstreamError:
      description: 上游不可用
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
```